

Lab Program 4

Integrate Git with Docker for Source-Controlled Application Builds

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Step 1 : create project folder

```
mkdir program-4
cd program-4
```

Step 2 : Initialize Node project

```
npm init -y
```

Step 3 : Create index.js

```
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req, res) => {
  res.send("Hello from Git Docker Integration!");
});

app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

Step 4 : Install dependency

```
npm i express
```

Step 5 : Create Dockerfile

```
FROM node:18 AS build
WORKDIR /app
COPY package.json ./
RUN npm install
COPY . .

FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]
```

Step 6 : Initialize Git locally

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
```

Step 7 : Create repo on github

```
git remote add origin https://github.com/<username>/<repo>.git
git push -u origin main
```

Step 8 : Connect local repo to Github

```
git remote add origin https://github.com/<username>/<repo>.git
git push -u origin main
```

Step 9 : Build Docker image from local source

```
docker build -t git-docker-app .
```

Step 10 : Run Docker container

```
docker run -p 3000:3000 git-docker-app
```

Open browser: <http://localhost:3000>

Step 11 : Verify

Verify on your Github account

```
> git remote add origin git@github.com:hvardhan1024/p4_test.git
> git push
fatal: The current branch master has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin master
```

To have this happen automatically for branches without a tracking upstream, see 'push.autoSetupRemote' in 'git help config'.

```
> git push --set-upstream origin master
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 12 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 724.94 KiB
Total 719 (delta 143), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (143/143), done.
To github.com:hvardhan1024/p4_test.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```